

EMMA M. GORIN, MSPH

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Education

Columbia University Mailman School of Public Health

Expected December 2026

PhD candidate in Environmental Health Sciences | Advisor: Dr. Jeffrey Shaman

- Dissertation research focuses on assessing the benefit of wastewater surveillance data for forecasting COVID-19 hospitalizations in New York State using a metapopulation model paired with an Ensemble Adjustment Kalman Filter (EAKF) for inference and forecasting.
- Selected coursework: Data Science; Atmospheric & Climate Science for Public Health; Emerging Infectious Diseases; Ordinary Differential Equations; Virology; Advanced Analytic Methods in Environmental Health Sciences; Bayesian Statistics for the Social Sciences; Intro to Computing for Engineers and Applied Scientists; Environmental Epidemiology; Epidemic Modeling Methods for Infectious Disease
- GPA: 3.9

Johns Hopkins Bloomberg School of Public Health, Baltimore, MD

May 2019

Master of Science in Public Health, Global Disease Epidemiology and Control

- Selected coursework: Epidemiology of Infectious Diseases; Concepts and Methods in Infectious Disease Epidemiology; Epidemiologic Inference in Outbreak Investigations; Infectious Disease Dynamics: Theoretical and Computational Approaches; Infectious Diseases and Child Survival; Issues for Water and Sanitation in Tropical Environmental Health; Vaccine Development and Application; Global Disease Control Programs and Policy; Epidemiological Methods (3 course series); Statistical Methods (4 course series); Spatial Analysis: ArcGIS; Spatial Statistics; Causal Inference in Medicine and Public Health; Stata Programming; Intro to SAS
- GPA: 4.0

Wesleyan University, Middletown, CT

May 2012

BA in Philosophy

- Semester abroad with IFSA Butler in Mérida, Mexico, Fall 2010
- GPA: 3.7

Other selected training:

- Summer Institute in Statistics and Modeling in Infectious Diseases, *University of Washington* Jul 2022
- Essentials of Teaching and Learning series, *Columbia Center for Teaching and Learning* Jan 2021
- REDCap 101 and REDCap 102, *Children's National Medical Center* Feb 2017
- Fundamentals of GIS, *University of California, Davis through Coursera* Jul 2016
- Supervising Sexual Health Educators, *ACT4Youth* Jan 2015
- Training of Trainers: Building Your Group Facilitation Skills, *Planned Parenthood NYC* Oct 2013

Experience

Graduate Research Assistant

Sep 2020 – Present

Columbia University Department of Environmental Health Sciences, New York, NY

Teaching fellow for "Molecular Epidemiology" and "Biomarkers: Quantitative Methods"

Jan-May and Sept-Dec 2023

Columbia University Department of Environmental Health Sciences, New York, NY

Teaching fellow for "Atmospheric and Climate Science for Public Health"

Sept-Dec 2021 and 2022

Columbia University Department of Environmental Health Sciences, New York, NY

Case Investigator/Contact Tracer I May – Aug 2020
New York City Health and Hospitals Test and Trace Program, New York, NY

Data Science Specialist Feb – Apr 2020
Johns Hopkins School of Public Health Department of Environmental Health and Engineering, Baltimore, MD

Private Stata Tutor Apr – May 2019
Baltimore, MD

Research Assistant Mar 2018 – Mar 2019
Key Populations Program, Johns Hopkins School of Public Health Department of Epidemiology, Baltimore, MD and Yaoundé, Cameroon

Clinical Research Coordinator Oct 2016 – Jul 2017
IMPACT DC, Children's National Medical Center, Washington, DC

Princeton in Latin America Fellow Jun 2015 – Jun 2016
Hospitalito Atitlán, Santiago Atitlán, Guatemala

Teen Pregnancy Prevention Program Co-Coordinator Jan – May 2015
Committee for Hispanic Children and Families, New York, NY

Teen Pregnancy Prevention Program Health Educator Sep 2012 – Jan 2015
Committee for Hispanic Children and Families, New York, NY

Intern Jun – Aug 2011
Brazos Abiertos, Mérida, Mexico

Awards and Fellowships

F31 Training Fellowship 2024–2026
National Institute of Allergy and Infectious Disease

Global Health Established Field Placement (GHEFP) Award 2018
Johns Hopkins Center for Global Health

Princeton in Latin America Fellowship 2015

Honors in Philosophy 2012
Wesleyan University

Publications, Presentations and Service

Publications

Gorin EM, Lyons CE, Jarrett BA, Djalo MA, Barreto K, Drame FM, *et al.* Mobility and HIV vulnerabilities among female sex workers in Guinea-Bissau: findings from an integrated bio-behavioral survey. *BMC Public Health* 2023;**23**:1856. <https://doi.org/10.1186/s12889-023-16744-y>.

Exum NG, **Gorin EM**, Sadhu G, Khanna A, Schwab KJ. Evaluating the declarations of open defecation free status under the Swachh Bharat ('Clean India') Mission: repeated cross-sectional surveys in Rajasthan, India. *BMJ Global Health* 2020;**5**:e002277. <https://doi.org/10.1136/bmjgh-2019-002277>.

Footer K, Windle M, Ferguson L, Hatcher J, Lyons C, **Gorin E**, *et al.* A Meta-Narrative Literature Synthesis and Framework to Guide Future Evaluation of Legal Empowerment Interventions. *Health Hum Rights* 2018;**20**:65–84.

Posters and presentations

Gorin E, Pei S, Yamana T, Larsen D, Shaman J. Retrospective Forecasting of COVID-19 Hospitalizations in New York State: Does Wastewater Data Improve Forecast Performance? Poster Session. Epidemics 10, 2025

Gorin E, Pei S, Yamana T, Larsen D, Shaman J. Dynamical Forecasting of COVID-19 in NYS With and Without Wastewater Data. Poster Session. 2025 Wastewater Surveillance Summit, October 2025

Gorin E, Larsen D, Shaman J. Approaches to using wastewater surveillance data for SARS-CoV-2 dynamical modeling and forecasting in NYS. Poster Session. Epidemics 9, December 2023

Gorin, E. Risk-set matching to assess the impact of bacterial infections on length of stay among COVID-19 inpatients in NYC hospitals. Presentation at Environmental Health Sciences weekly seminar series, November 2021

Gorin, E. COVID-19 in NYC: Modeling transmission dynamics using traffic as a proxy for mobility. Presentation at Environmental Health Sciences weekly seminar series, March 2021

Gorin, E. HIV programming for key populations in Cameroon and mobility among female sex workers in Guinea-Bissau. Poster session. Global Health Day at Johns Hopkins Bloomberg School of Public Health, April 2019

Peer review service

BMC Infectious Diseases

Social Sciences & Humanities Open

AIDS and Behavior

Skills

Languages: Native English, fluent Spanish, conversational French

Software: R, MATLAB, Stata, Python, Claude, SAS, ArcGIS, REDCap, Power BI